

# **A Dual-Sensor UAS Approach to Wetland Restoration:**

Integrating Bathymetric and NIR LiDAR in the Mississippi River Valley

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## **Introduction & Background:**

Arkansas's bottomland hardwood forests are a corner stone of the Mississippi River Valley's ecological and economic health, yet they face a critical tipping point (Society of Wetland, 2025). While these wetlands provide an estimated \$37.4 billion a year in ecosystem services, which include flood mitigation, water filtration, and critical habitats for numerous species (Dogwood, 2017; Rose, 2026-e, Braddock, 2026-c), they have suffered an estimated 70% loss since 1780s here in Arkansas, mostly due to agricultural development (Environmental Law, 2024, p1). The Cache- Lower White River System, designated as a Wetland of International Importance by the Ramsar convention, home to over 247 bird species, 50 mammal species, and 48 species of amphibians and reptiles, contains the largest remaining expanse of this forest type, making its preservation a regional and international priority (society of wetland, 2025). However, effective restoration is currently hampered by the limitations of traditional mapping. Standard topographic LiDAR and satellite imagery fail to penetrate the water's surface, leaving a data gap where critical sediment build up and subsurface inundation remain unmapped (USDA, 2014).

This project challenges that gap by asking: How can high-resolution bathymetric LiDAR data accurately monitor changes in inundation extent and sediment dynamics within Arkansas's bottomland hardwood forests to prioritize restoration efforts? It is hypothesized that integrating a dual-sensor UAS system, utilizing a 532nm green laser for water penetration and a 1064nm near-infrared (NIR) laser for surface detection, will provide a seamless 3D model of the terrain, allowing for more precise flood risk modeling than traditional remote sensing techniques. By survey-mapping a critical 50-acre track as the initial location, this study will apply a spatial and sustainability perspective to identify dead zones of sediment buildup, providing the high-tech insights needed to protect Arkansas's remaining wetland heritage (Szafarczyk, 2022).

## **Literature Review:**

The conservation of Arkansas's bottomland hardwood forests has reached a critical intersection (Demas, 1996). Historically, these wetlands have acted as the primary hydrological regulators for the Mississippi Alluvial Valley, but as Dogwood Alliance mentions, over 70% of Arkansas's wetlands have been lost. The Cache-Lower White River System represents the largest remaining fragment of this ecosystem and is recognized by the Ramsar Convention as a wetland

of international importance (Society of Wetland, 2025). Beyond their ecological status, these wetlands provide nearly \$37.4 billion in ecosystem services, serving as vital infrastructure for flood mitigation and water filtration. However, as noted by the Environmental Law Institute, recent shifts in the regulatory status of “non-WOTUS” wetlands make high-precision mapping essential for navigating new conservation and legal landscapes (2024).

Despite the importance of these areas, traditional remote sensing techniques struggle to provide the granular data needed for effective restoration. Standard topographic LiDAR utilizes a 1064nm NIR laser, which is highly effective for forest canopy and dry topography but is absorbed by water, rendering it incapable of mapping submerged terrain. This creates a data gap in shallow water environments where sediment accumulation often occurs. In contrast, bathymetric LiDAR utilizes a 532nm green-spectrum laser that can penetrate the water column (He et. al., 2024, Szafarczyk, 2022). This technology allows for the creation of a seamless digital elevation model (DEM) that transitions from the forest floor to the riverbed within the data voids typical of NIR-only systems. Research by Anokye indicates that integrating LiDAR with multispectral data increases the accuracy of understanding wetland flood dynamics by capturing both the structural and spectral signatures (2026-b). However, integrating these data types can be time consuming and costly, like Huizinga’s project “Use of High-Resolution Topo-Bathymetry to Assess the Shoreline Topography and Future Development of a Slackwater Harbor near Dardanelle, Arkansas” in October of 2021, where they merged multi-beam sonar with UAS gathered NIR LiDAR, or for the combination of LiDAR and sonar used to guide the “Master Plan for Development and Management of Blue Mountain Lake” in 2024.

Though, as Legleiter et. al. explains in their paper, “Remote Sensing of River Bathymetry: Evaluating a Range of Sensors, Platforms, and Algorithms on the Upper Sacramento River, California, USA” in 2018, there are a range of ways to explore bathymetry, each with their own limits. When it comes to multispectral imaging, either satellite or UAS, the depth estimation of the water is contingent upon the turbidity/color/clarity of the water and can be less accurate than other forms, and it is often hindered by tree canopies or atmospheric interference. While traditional sonar techniques are incredibly beneficial for bathymetry, they are often hindered by access to navigable water, making the complete surveying of shallow wetlands or forested areas impossible. Legleiter stresses that airborne bathymetry is the only viable

method for mapping non-navigable waters. By bridging the gap between terrestrial and sub-aquatic mapping, this project leverages cutting edge remote sensing to identify sediment in dead zones, areas where historical drainage patterns have been disrupted, to ensure the long-term sustainability of the Cache-White River's remaining hardwood forests.

## **Methodology:**

To appraise the inundation and sediment dynamics of the Cache-Lower White River System, this study employs a high-resolution, multi-sensor approach using an Unmanned Aircraft System (UAS). This platform allows for lower-altitude flights than traditional manned aircraft, resulting in the high point density required to penetrate dense bottomland hardwood canopies (Anokye, 2026-b). The primary instrument to be used in this project is a dual-mounted LiDAR system consisting of a 532nm green laser and a 1064nm NIR laser. Following the technical standards identified, the 532nm laser is utilized for its unique ability to penetrate the water column, while the 1062nm laser provides a high intensity return from the water surface and terrestrial vegetation (He, 2024; Legleiter, 2019). This dual-pulse approach ensures a seamless transition between the submerged aquatic and surround topography, eliminating the data gap at the land-water interference common in single sensor surveys.

This proposed survey would be conducted over a designated high-priority 50 acre-track using a dual-grid pattern. Flights will be conducted in parallel lines at consistent altitude to maintain a uniform footprint. To minimize any canopy shadows, the UAS will fly perpendicular paths: first from North to South, that is followed by a second grid traveling East to West. A 60% side-lap between flight paths will be maintained to ensure every point on the forest floor is viewed from multiple angles, significantly increasing the probability of ground returns within the hardwood canopy (USDA, 2014; Huizinga et al., 2021; U.S. Army Corps, 2024).

This project focuses on a spatial and quantitative framework, as well as a human-environment and sustainability perspective that heavily affects the post processing phase. First, the point clouds will be filtered to produce a DEM and Bathymetric model to produce a Topobathymetric Digital Elevation Model (TBDEM). Subtracting the submerged terrain from the water surface returns will allow for the calculation of exact water storage volumes and sediment rates. By identifying the dead zones, wherein sediment accumulation has hindered the natural

flow of water, the data will be layered with AGFC land-use maps to determine how upstream runoff correlates with downstream habitat degradation. To ensure the model's legitimacy, five survey-grade Ground Control Points will be established using RTK-GNSS within the 50-acre study zone. These points will serve as the anchor to which the LiDAR point cloud is georeferenced, ensuring the sub-decimeter accuracy necessary for precise hydrological modeling.

### **Budget and Cost-Benefit Analysis:**

The costs associated with this project are estimated based off information from both *E38 Survey Solutions* and *Airship Photography and Drone Services*. Wherein the total estimated costs for this project range from \$145,000 to \$165,000. The largest cost associated with this project comes from purchasing the hardware and equipment, focusing on research-grade equipment: the sensor suite, UAS platform, and RTK-GNSS combined will be between \$120,000 and \$135,000. Other costs will include field operations and labor, concerning the designation of ground points, and survey operations, with an estimation of a 3-5 day survey to cover the 50 acres will cost between \$10,000 and \$15,000. Finally, the costs for processing the data and analysis could range anywhere from \$5,000 to \$15,000, depending on software licensing and storage capacity.

However, while the upfront costs are higher than many other remote sensing techniques, the return on investment is justified, especially once you factor in the protection of \$37 billion in ecosystem services this technology can provide. Traditional boat mounted sonar for this location would be impossible due to the shallow debris, and regular LiDAR would provide zero bathymetric data. This budget allows for the one-time purchasing of equipment that could provide countless scientific benefits to Arkansas's ecosystems.

### **Expected Outcomes and Conservation Impacts:**

This integration of dual pulse LiDAR within the Cache-Lower White River System represents a critical shift to proactive wetland management. By bridging the terrestrial-aquatic data gap, this project aims to provide the first seamless 3D inventory of Arkansas's most vital bottomland hardwood forests. The primary output for this study will include a high-resolution DEM and TBDEM that identifies the exact morphology of the 50 acre site, expected to help identify sediment dead zones. Furthermore, the data will produce precise inundation maps that

simulate how floodwater moves through the forest between terrestrial, surface water, and submerged terrain, and by providing a volumetric analysis of water storage capacity that is currently unavailable to state agencies.

This project is rooted in the immense value of the ecosystem services at stake. With an estimated \$37.4 billion in services provided by the remaining 30% of Arkansas's wetlands, the cost of data blindness is simply too high. Traditional remote sensing techniques often fail to see the sediment that is killing these forests from the bottom up. By utilizing the 532nm green laser to map non-navigable portions of this system, we can provide Arkansas with a surgical tool for restoration. Instead of relying on broad conservation efforts, this data will allow prioritization of critical areas. Identifying where dredging, levee breaching, or rerouting will yield the highest returns on investment. This project can serve as a scalable pilot to ensure that the largest remaining expanse of bottomland hardwood forests within the Mississippi River Valley can be preserved for future generations, transforming the ways in which Arkansas monitors and maintains its critical wetland infrastructure.

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